



MINING DEALER BEST PRACTICE

Salvage C175 Engine Block Coolant Passage Using Metal Spray

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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December 2023
4.02-231220-1

1.0 Introduction

Significant cost savings can be achieved by salvaging the coolant passages on the top deck surface of the C175 engine cylinder block using metal spray technology.

2.0 Best Practice Description

Coolant passage salvage consists of machining the damaged surface using an end mill to eliminate the pitting, grit blasting the surface to create a bond interface, metal spraying the surface, and machining the top deck to create a uniform surface.



Figure 1. Coolant passage’s surface affected by pitting

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3.0 Implementation Steps

3.1. Rough cut: Eliminate the pitting on the coolant passages surface using an end mill to machine a pocket that contains the damage. Most of the time a 0.040" deep pocket is enough for removing pitting.



Figure 1 Machined pocket on coolant passage surface.

3.2. Top deck bore and thread protection: Use anti-bond materials to cover and protect all the threads and bores which are near to the area to be sprayed on the top deck.



Figure 2. Top deck bore and thread protected

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3.3. Grit blasting: Follow the instructions in SEBF2120 “Thermal Spray Procedures for Engine Cylinder Block Top Surface”. Propel a stream of aluminum oxide against the block surface under high pressure to prepare the work piece for metal spray.



Figure 3. Grit blasted surface

3.4. Metal spraying the surface: Using an industrial robot and metal spray equipment, it is possible to control velocity, distance and position of the spray with a high degree of accuracy improving the strength of the bond and finish of the coating. For the spraying parameters use the specifications found in SEBF2120 “Thermal Spray Procedures for Engine Cylinder Block Top Surface”.

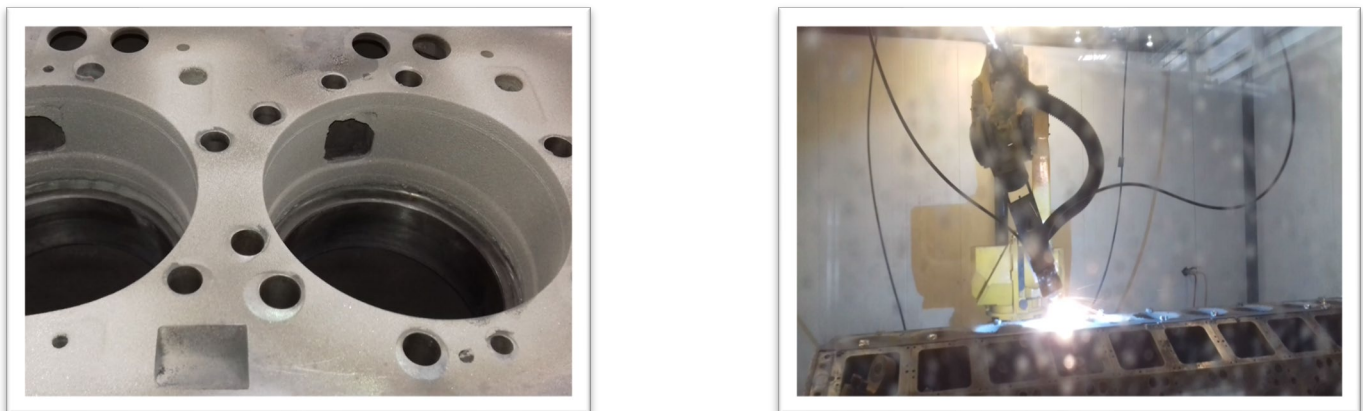


Figure 4. Sprayed coolant passages using an industrial robot

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3.5. Finishing cut: Machine the top deck surface according to the specifications found in the Reuse and Salvage Guidelines SEBF9185 “Specifications and Salvage for Cylinder Blocks in C175 Engines”.

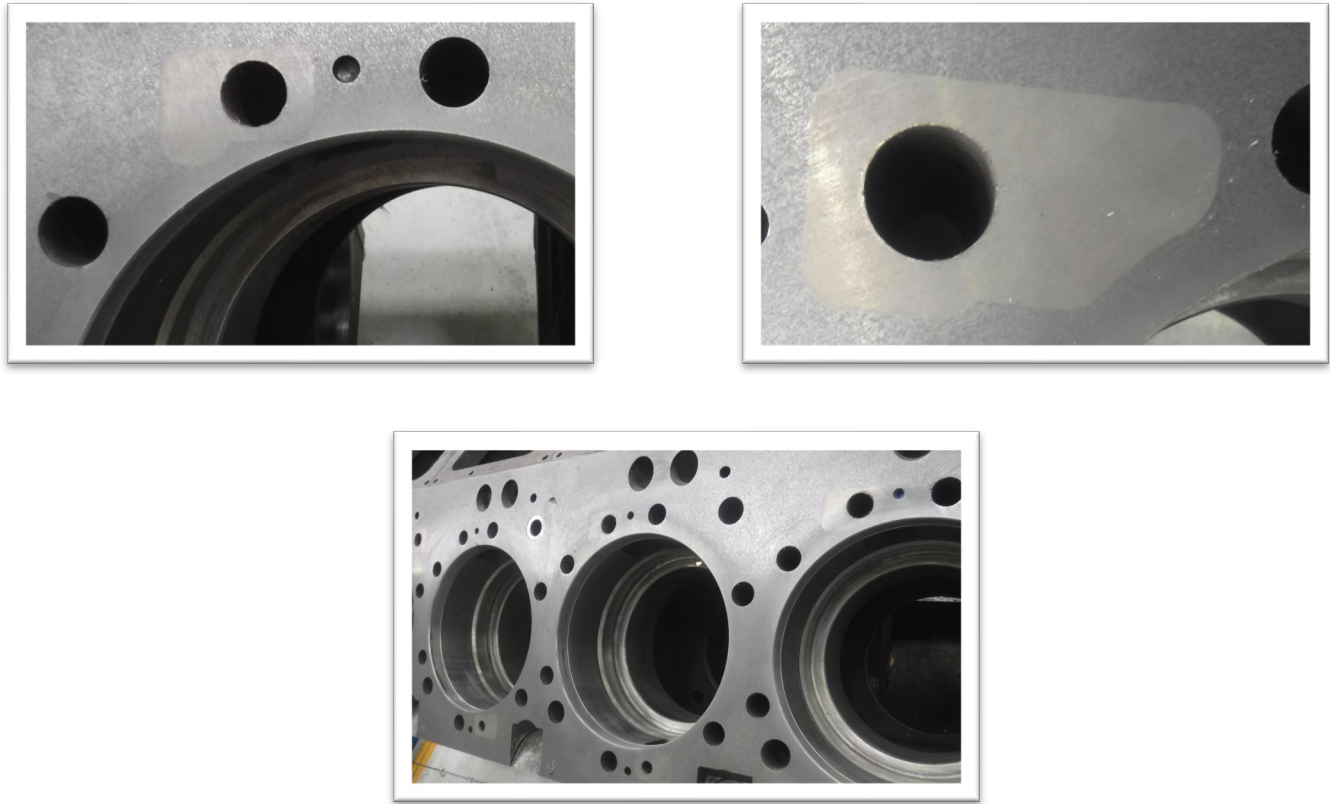


Figure 5. Machined top deck surface

4.0 Benefits

- 4.1. Increases the life of the salvage by using wear resistant materials to create the coating.
- 4.2. Eliminates the need to metal spray the whole top deck surface when pitting exists on coolant passages.
- 4.3. This salvage process is cheaper than spraying the whole top deck surface and saves valuable consumables and reduces labor time as only the areas affected by pitting are sprayed.

5.0 Resources Required

- 5.1. Multi-purpose CNC machining center (Rottler F100 Series or similar)
- 5.2. Grit blasting equipment
- 5.3 Industrial robot
- 5.4 Metal spray equipment

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6.0 Supporting Attachments / References

- 6.1. SEBF9185 “Specifications and Salvage for Cylinder Blocks in C175 Engines”
- 6.2. SEBF2120 “Thermal Spray Procedures for Engine Cylinder Block Top Surface”
- 6.2. SEBF9238 “Fundamentals of Arc Spray for Reconditioning Components”

7.0 Related Best Practices

- 7.1. Mounting Fixture for Arc Spray Gun
- 7.2. Setting-Up the Robot Using a Cone Tool
- 7.3. Salvage C175 Engine Cylinder Block Upper and Lower Liner Bores Using Metal Spray

8.0 Acknowledgements

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